

## Data Card — Latent Probing of Toxicity Detection

### Dataset Description

The training data combines four publicly available datasets to improve the generalization of a linear toxicity probe across different forms and sources of toxic language.

ToxicChat: 768 examples (384 safe, 384 toxic)

RealToxicityPrompts: 536 balanced examples

BeaverTails: 200 examples (100 safe, 100 toxic)

Jigsaw Toxic Comment Classification: 800 examples (400 non-toxic, 400 toxic)

The final training set contains 2,304 examples.

All examples were converted to a common binary classification format:

0 = Safe / Non-toxic

1 = Toxic

The text prompts/comments were passed through the required Google Gemma 2 2B model. Hidden representations were extracted using the competition's mean-pooling procedure and used as inputs to the toxicity probe.

Input vs. Output

Input: A text prompt or comment.

Output: A binary toxicity prediction.

0 = Safe / Non-toxic

1 = Toxic

The probe operates on the hidden-state representation produced by Gemma rather than directly on the raw text.

### Classification Problem

This is a binary toxicity classification task.

The objective is to determine whether a given text should be classified as toxic or non-toxic based on the semantic information encoded in the model's hidden representations.

Because the datasets originate from different sources and use different annotation schemes, their labels were mapped into a common binary format before training.

Labeling

Labels were derived from the original annotations provided by each dataset.

ToxicChat: The dataset's toxicity/safety labels were converted to the competition's binary format.

RealToxicityPrompts: Examples were selected using the dataset's toxicity scores and converted into safe/toxic classes.

BeaverTails: The dataset's safety annotations were mapped to safe/toxic labels.

Jigsaw Toxic Comments: Comments were classified as toxic when the dataset's toxic label indicated toxicity; otherwise they were treated as non-toxic.

Where necessary, balanced subsets were selected to reduce class imbalance in the training data.

### Data Creation

No new user-generated or personally collected data was created.

The training set was constructed by selecting subsets from publicly available datasets. ToxicChat, BeaverTails, and Jigsaw were sampled to maintain balanced safe/non-toxic and toxic classes. The Jigsaw subset consisted of randomly sampled and shuffled examples rather than manually selecting individual toxicity subcategories. RealToxicityPrompts was used to construct balanced training and held-out evaluation subsets based on its toxicity scores.

The final dataset was then converted into Gemma hidden-state representations using the same preprocessing and mean-pooling procedure used for the competition.

#### Web Collection Ethics

The datasets used in this project were obtained from publicly available research datasets rather than collected directly from individuals.

Because the data contains potentially offensive, abusive, hateful, or otherwise toxic language, it may contain material that is disturbing or inappropriate for some users. The project does not attempt to identify individuals or infer sensitive personal characteristics. The data is used solely for research and experimentation in toxicity detection and mechanistic interpretability.

Potential limitations include annotation subjectivity, demographic and linguistic biases, differences in labeling standards between datasets, and the possibility of false positives or false negatives. These considerations are important because an imperfect toxicity classifier could incorrectly flag benign content or fail to detect harmful content.